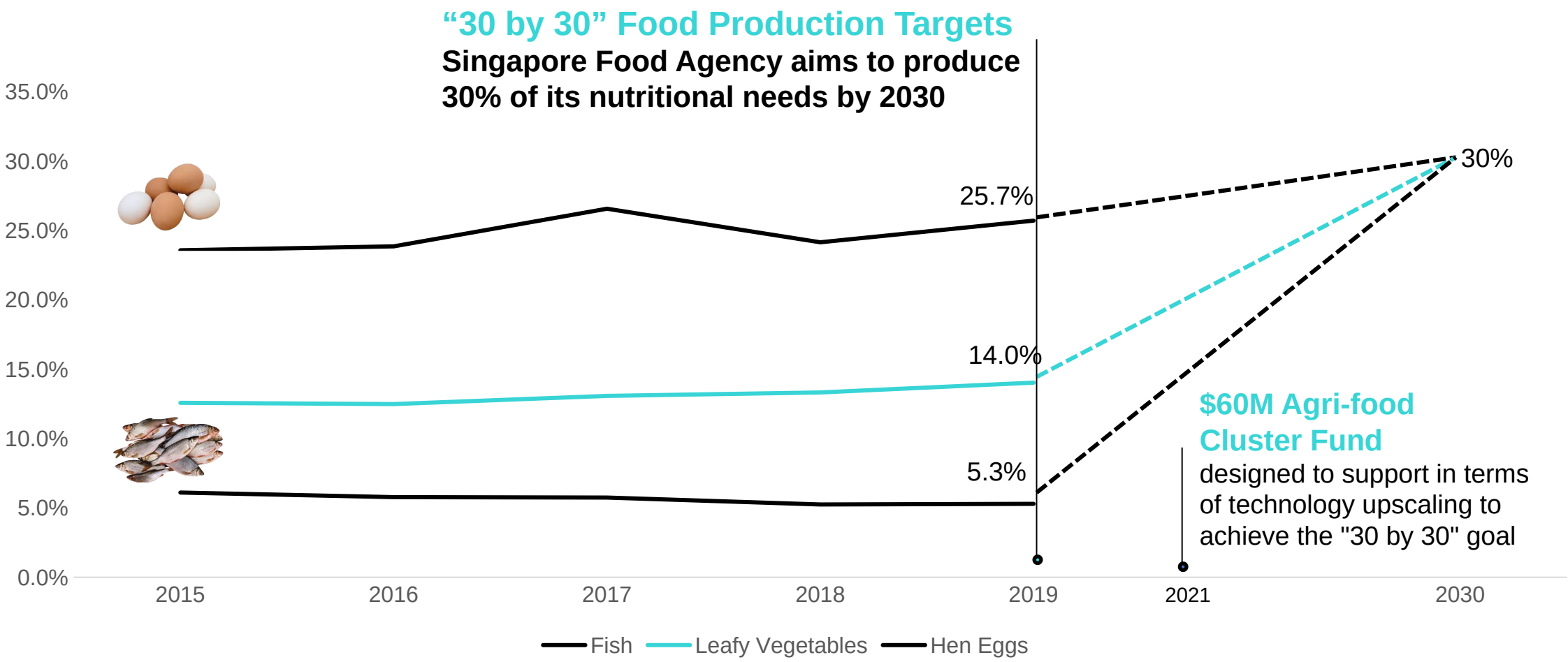


# AMBIENT SYSTEMS

Climate At Your Service

**Investment Deck Apr 2021**

# Local farms sees supports from local government sectors




*Agriculture sector of Singapore is engaged mainly in the production of eggs, fish and vegetables for local consumptions
(1) Fish only includes seafood from coastal and land-based farm production and excluding local landings of seafood (seafood catches from local waters)
Source: Agri-food and Veterinary Authority, Singapore Food Agency, National Parks Board

# Difficulties in replicating climate conditions in indoor farms

## OPTIMAL CONDITIONS DEPENDS ON SEVERAL FACTORS

**Day-time Temp**: 20-24C
**Night-time Temp**: 10-12C
**Humidity**: Low humidity
**(Crown):** 18C
**Light Intensity**: 800-1200 micro-mol/m2/s PPF



## FARMERS HAVE NO CONTROL ON BUILDING CONDITIONS

**No access to building data**

## QUALITY CROP ARE TIME-SENSITIVE

**Crop growth requires real-time monitoring**

## LACK OF DYNAMIC SOLUTION FOR OPTIMAL ENVIRONMENT

**No infrastructure integration with Air Conditioning and Machine Ventilation (ACMV)**

# Agri-tech remains key area of investment in Southeast Asia

## Farm Management Software, Sensing & IOT is the top 5 most invested category in the region (2019)

1. **In-Store Retail & Restaurant Tech**
   Shelf-stacking robots, 3D food printers, POS systems, food waste monitoring IoT.
2. **Online Restaurants and Meal Kits**
   Startups offering culinary meals and sending pre-portioned ingredients to cook at home.
3. **eGrocery**
   Online stores and marketplaces for sale & delivery of processed & un-processed ag products to consumer.
4. **Midstream Technologies**
   Food safety & traceability tech, logistics & transport, processing tech.
5. **Farm Management Software, Sensing & IoT**
   Ag data capturing devices, decision support softwa big data analytics.

**$423M**
**Invested**

**+33%**
**YoY Investment Growth**

**+41%**
**YoY Deal Growth**

Source: 2020 , Agfunder

Climate At Your Service

# Connected.
# Integrated.
# Automated.

ɔmprehensive climate management solution
that combines state-of-the-art hardware and
software technologies to deliver real-time
insights and advanced optimization models.

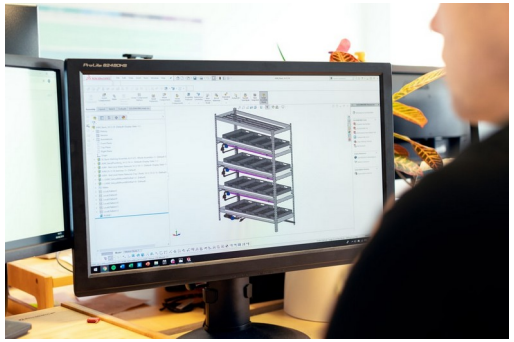
AMBIENT SYSTEMS
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# Our Features



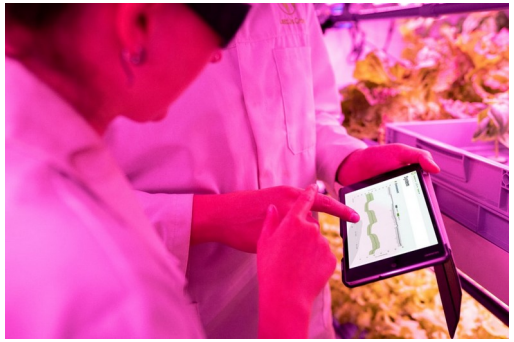
## Seamless Building Integration

Enjoy full connectivity within your farm and perform real-time environmental adjustments based on crop



## Monitoring In Real-Time

Access real-time data through online dashboards and receive push alerts anytime anywhere



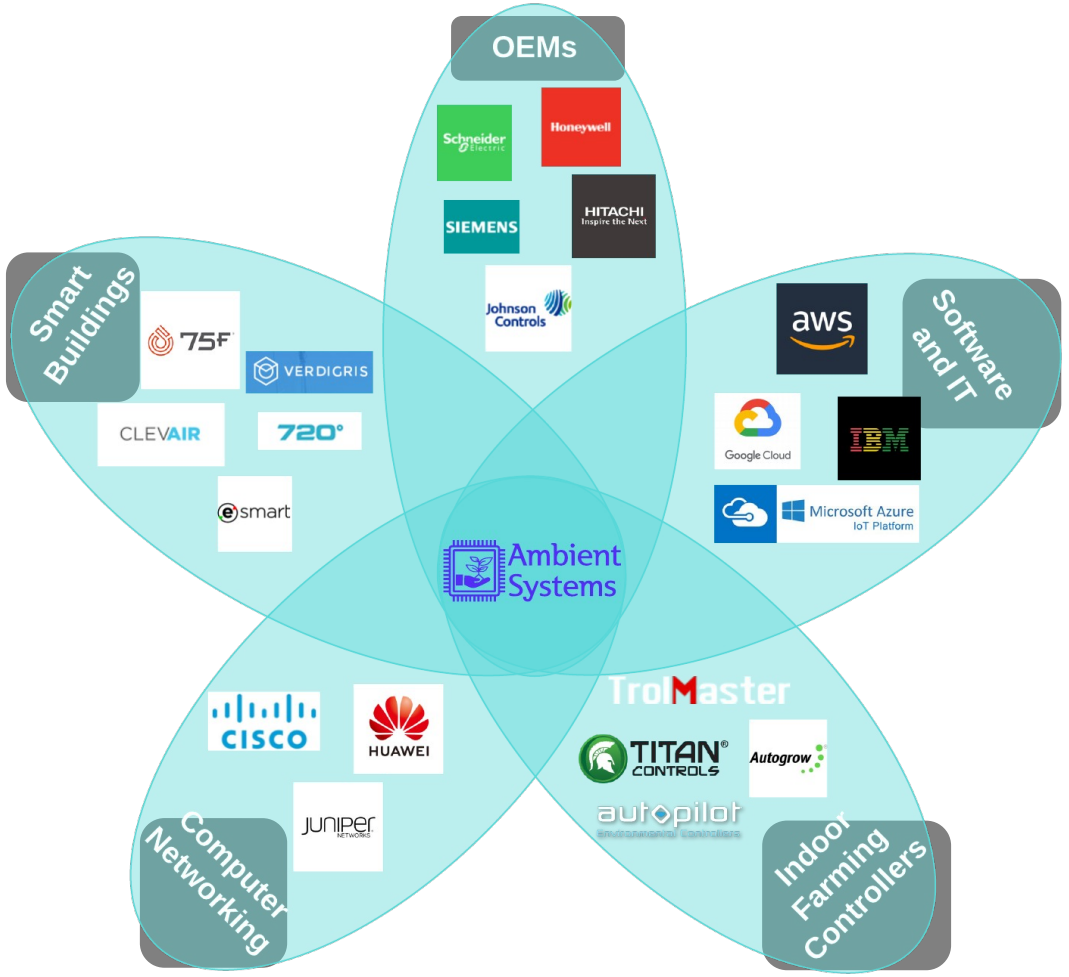
## Grow With Data Insights

Apply data-driven solutions to understand factors that affect growth and quality, enabling better crop health

# Video Demonstration

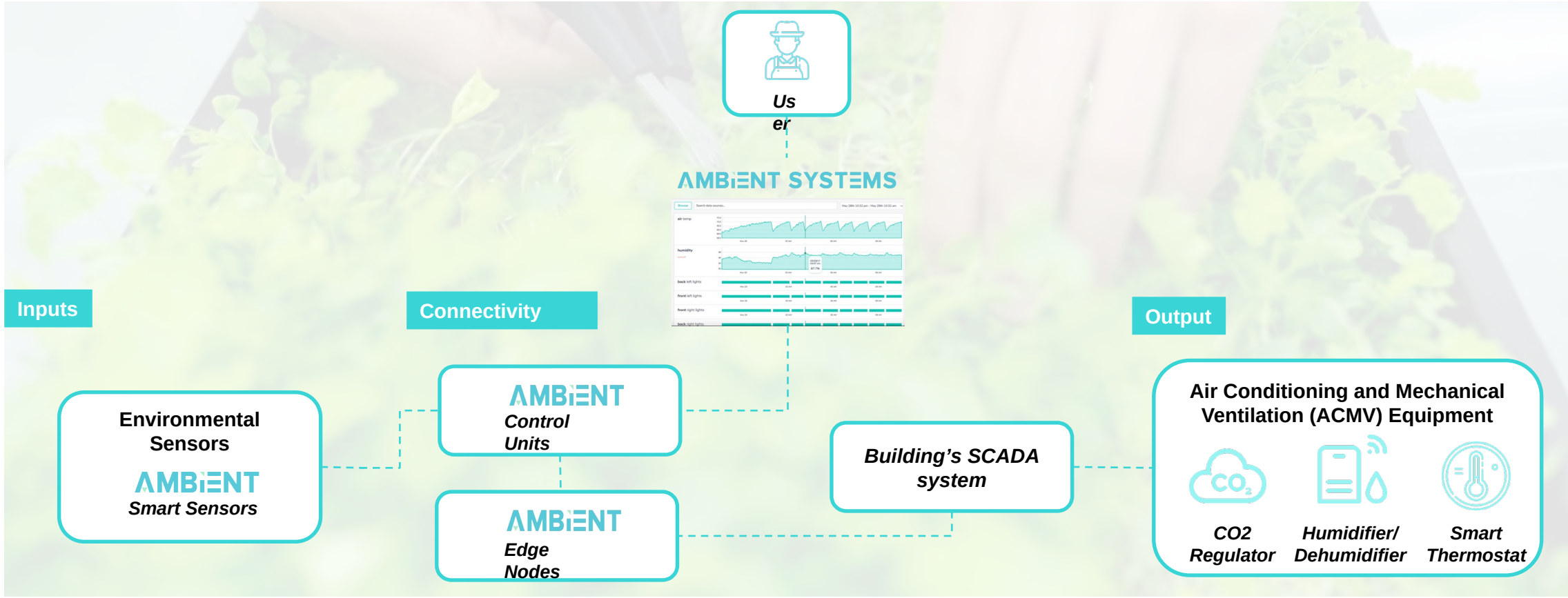
**[In Progress]**

# Consolidating Current Urban Farming Technologies into a Turn-Key Solution




**Ambient provides a tailored solution for the indoor farming beachhead market by positioning itself at the intersection of multiple segments.**

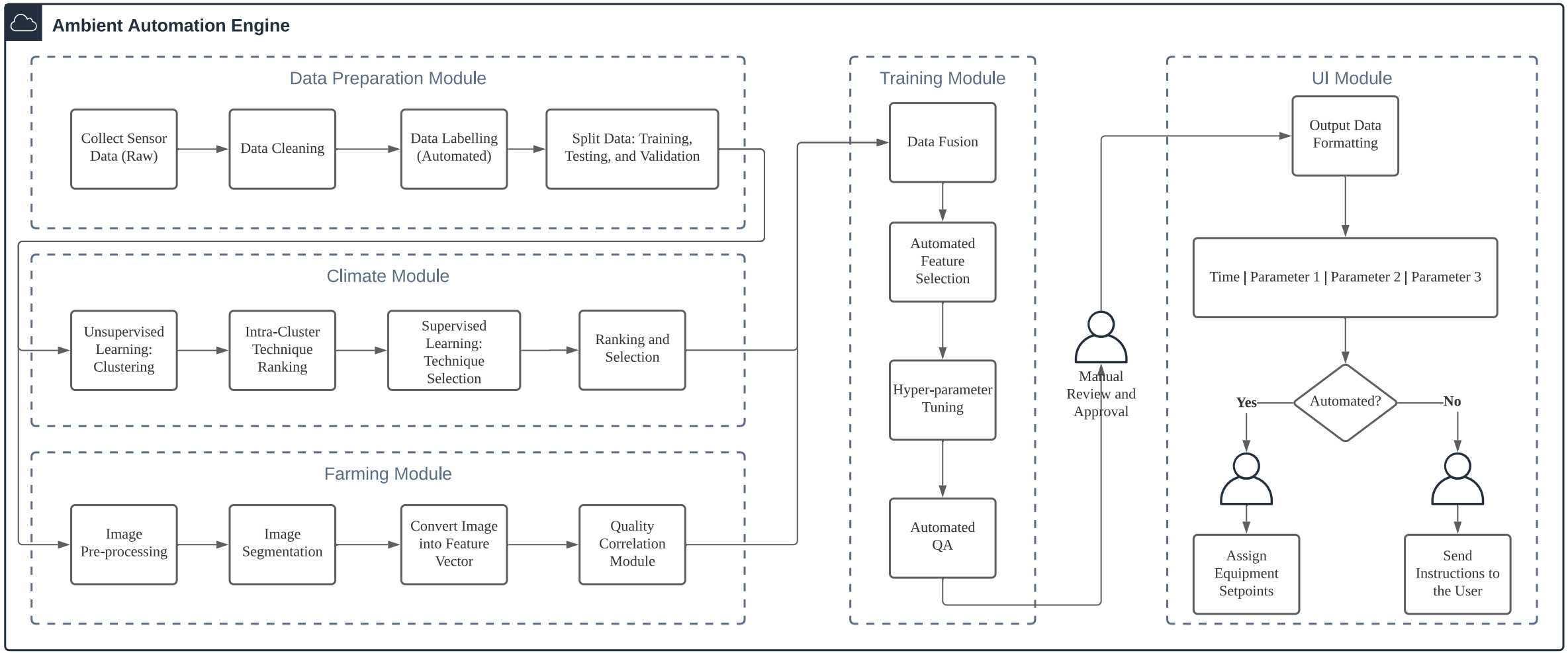
# Ambient Technology



# IP-enabled competitive advantage

1. **Patent**: Automated Climate Management for Indoor Farming Facilities
2. **Copyright**: Proprietary Database Design and Algorithms for the Automation Engine
3. **Trade Secret**: Operational Data on Creating an Optimal Indoor Climate

# Deep dive into the Machine Learning Optimization Model




## Patent focuses on climate optimization specific to indoor farming

# Value-based Pricing driven by our

# For Urban Farms

**l Pricing Table for 1 Small Sized Farm**
(220sqft or 20.4m2)

| SGD($) | | | |
|---|---|---|---|
| **Implementation Cost** | Hardware | **Quotation** | IoT Sensors<br>Device & Installation<br>Integration |
| | Installation | **Quotation** | **One-time cost** |
| **Operational Cost** | Monthly Data Management | **$1,000.00** | **Recurring cost** |
| | Monthly System Maintenance | **$500.00** | Higher Crop Yield<br>Higher Quality<br>Harvest Time Prediction |
| **Use case specific** | Monthly Estimated (% of Crop Value) | **5%** | |

# Our Financial Projections

| Base | Y1 | Y2 | Y3 | Y4 | Y5 |
|---|---|---|---|---|---|
| **Number of farms managed** | 14 | 45 | 141 | 354 | 566 |
| **Area of Farm Size Managed (m2)** | 1,164 | 3653 | 11,466 | 28,873 | 46,226 |
| **No. of Container Farms Equivalent** | 57 | 179 | 562 | 1415 | 2266 |

| | Bear | Base | Bull |
|---|---|---|---|
| **Runway (Months)** | 20 | 17 | 15 |
| **Funding Gap (S$)** | $510K | **$455K** | $415K |

*Software Subscriptions (SGD$)*

| | | | | | |
|---|---|---|---|---|---|
| **Revenue** | $302,069 | $1,004,417 | $3,351,733 | $9,768,693 | $20,031,648 |
| **Gross Margin** | 91.15% | 91.65% | 92.14% | 92.63% | 93.10% |
| **P&L** | ($402,662) | $94,621 | $1,910,903 | $7,095,040 | $15,262,342 |
| | ($402,662) | ($308,041) | $1,602,862 | $8,697,902 | $23,960,243 |
| **Discount Rate** | 10% | | | | |
| **Net Present Value** | **$21,401,829** | | | | |

# Pilot Project

**[In Progress]**

# Ambient's Climate As A Service solution can be seamlessly deployed in any industries that require a controlled environment












# About Us

**Ambient is a deep-tech spin-off from the National University of Singapore that is developing an integrated hardware-software solution for climate management of vertical farms.**

**The product leverages intellectual property on securely combining various sources of data and machine learning to enhance building performance.**

*Product Development*

- **MComp, Computer Science**
  *National University of Singapore*
- **BASc, Civil Engineering, Sustainable Energy**
  *University of Toronto*
- **Serial Entrepreneur**
  5 years of experience in cleantech & deep tech start-ups (YC-backed)

*Corporate Development*

- **BBM, Finance**
  *Singapore Management University*
- **Strategy and Fundraising**
  3 years of experience across e-commerce, gaming and payments



# AMBIENT SYSTEMS

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**Contact: ivan@ambientsystems.ai**
**ambientsystems.ai**

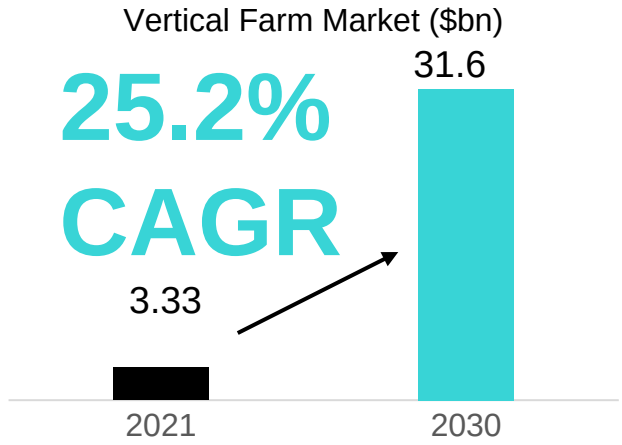
# Appendix

# Global Vertical Farming Market poised for double-digit growth

By 2030, **60%**
of the world's population
will live in urban cities



Agriculture Connectivity
could unlock more than
**$500 billion**
in GDP by 2030



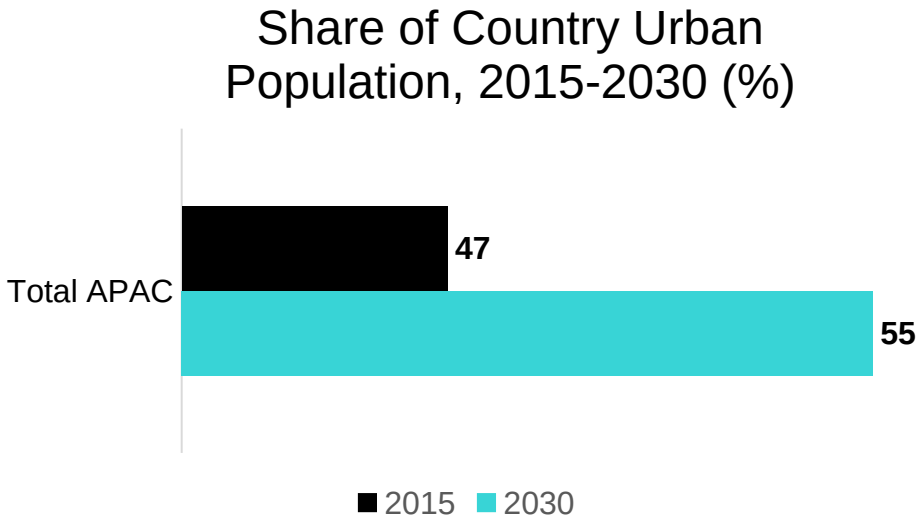

Source: Precedence Research, Mckinsey

*Private and Confidential*

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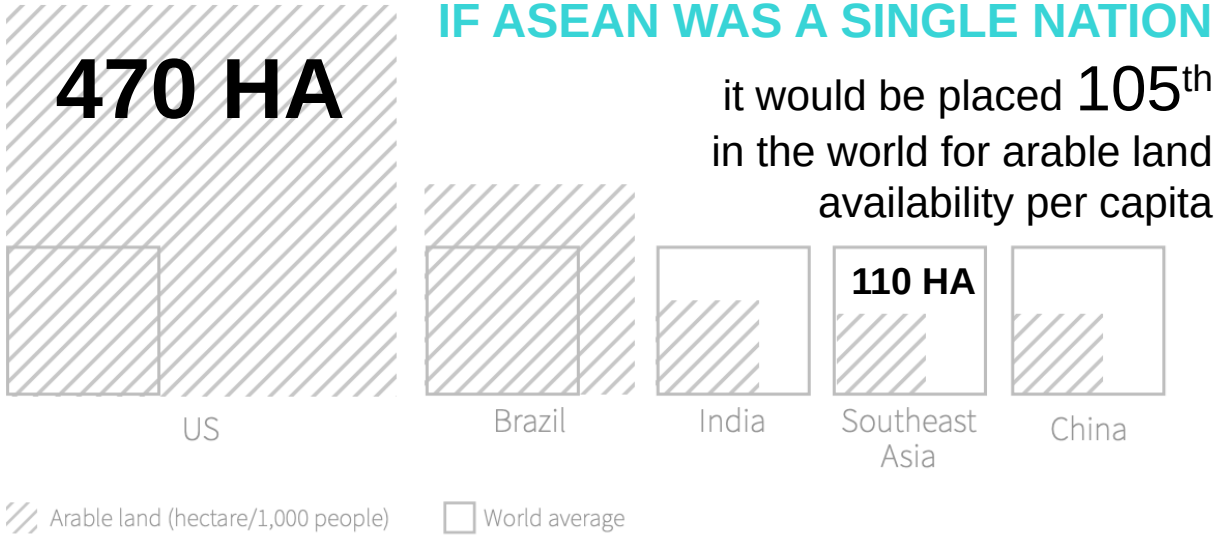
# Challenges of feeding the increasing population in Urban Asia Pacific

## Increasing Urban Population



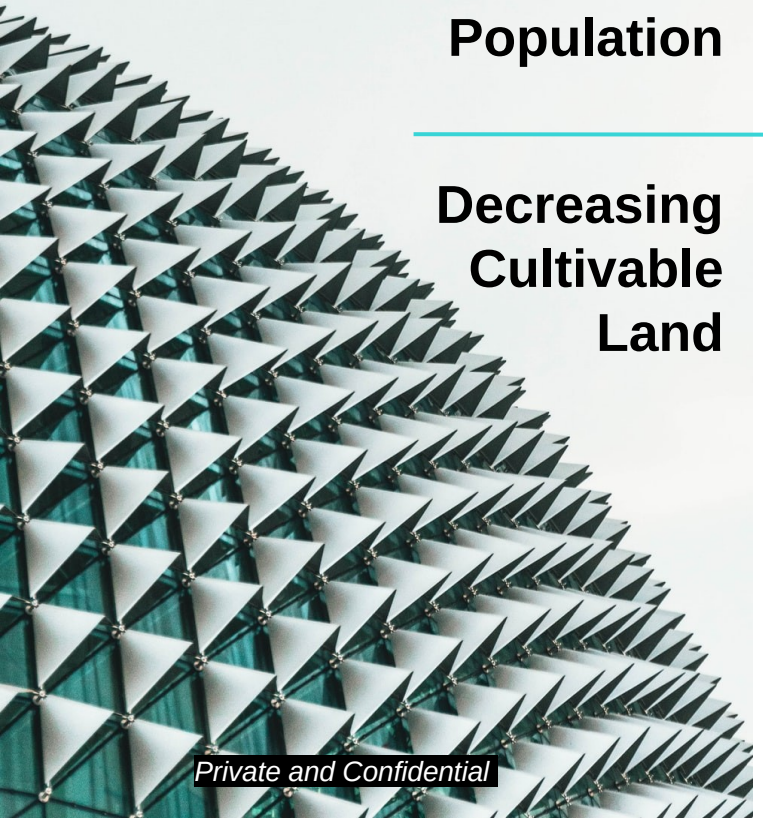

**555 MILLION**
people will be added to urban cities by 2030

## Decreasing Cultivable Land




**IF ASEAN WAS A SINGLE NATION**

it would be placed 105th in the world for arable land availability per capita

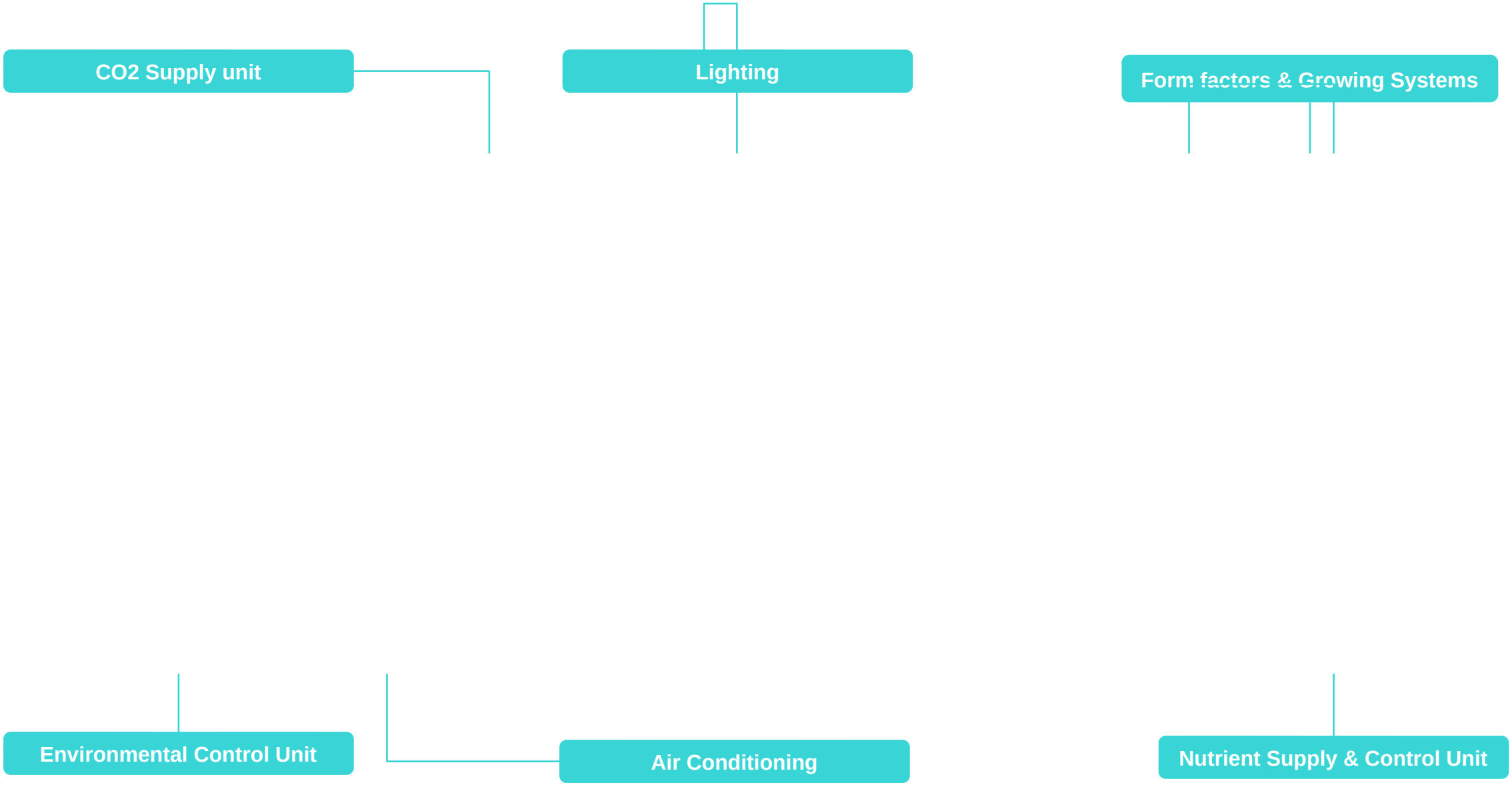


*Private and Confidential*



Source: Food and Agriculture Organization of the United Nations (2017), PWC (2020)

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# [Insert Revised Business Model]

**Key Partners**

NUS GRIP
Farm Operato
Building Owne
HVAC & BMS Ver
Government Agen

**Key Activities**

ements Gathering
rm Development
tem Installation
model design,
ment, training, and
tuning

**Key Resources**

IP
Platform
ML Model

**Value Proposition**

Initial: Enable successful climate management of vertical farms by measuring key parameters and ensuring process repeatability for future harvests.

data: increase yield, o
rgy use, and reduce la
costs.

**Customer Relationships**

Tailored Pricing
Modern Digital Platform
Personalized Support

**Channels**

Direct Outreach
Equipment Suppliers
Farmer Cooperatives

**Customer Segments**

Indoor Farms

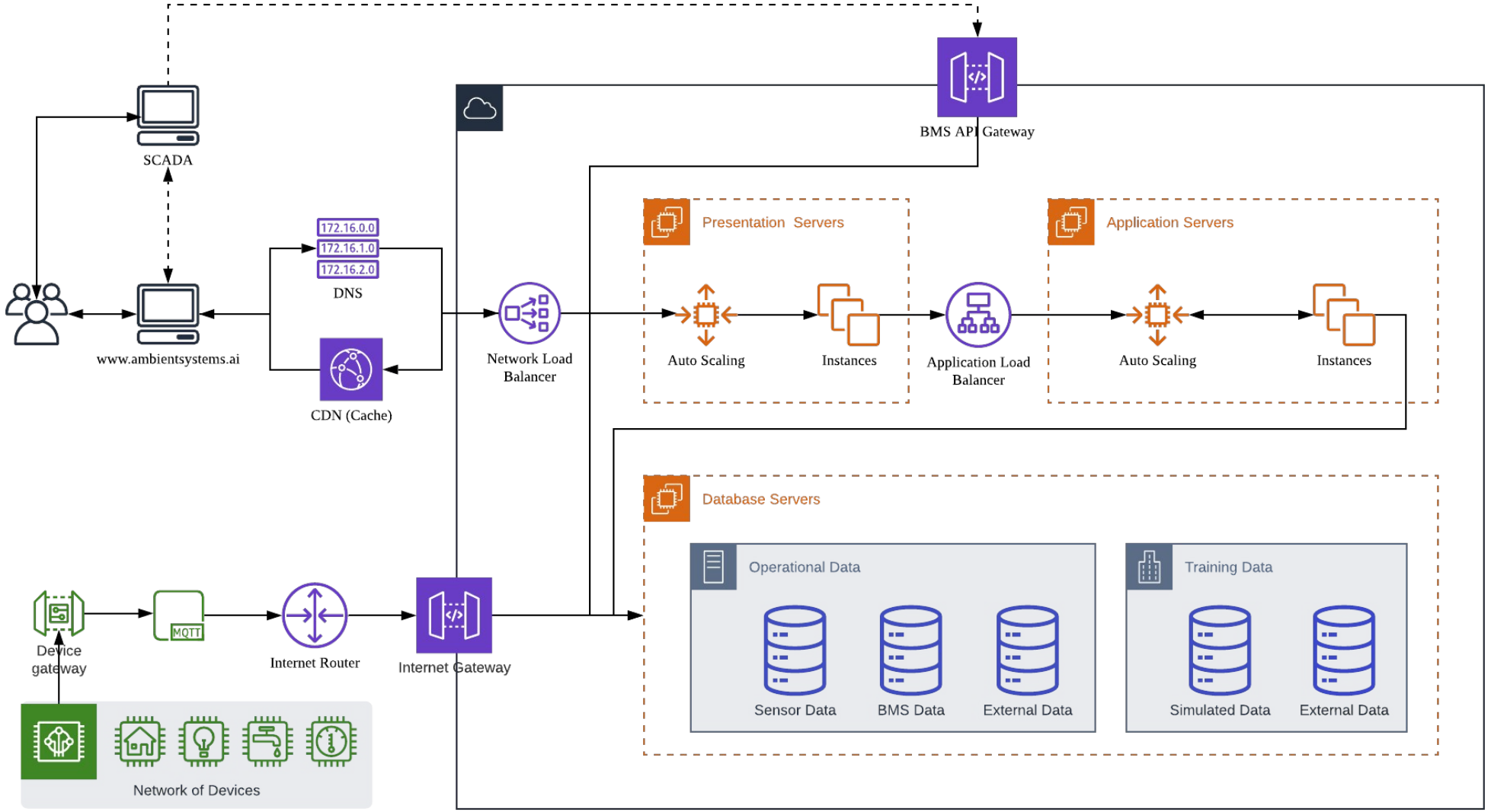
**Cost Structure**

**Revenue Streams**



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# [Insert Revised Technology Architecture]



# We offer end-to-end climate control solutions that are guaranteed with our Pay As You Grow revenue sharing model

| | AMBIENT SYSTEMS Climate At Your Service | Autogrow | 75F | Johnson Controls |
|---|---|---|---|---|
| **Segment** | **Multiple** | **Indoor Farming Controller** | **Smart Buildings** | **OEM** |
| **Proximity to Customer (Low Logistics Cost)** | ✓ Yes | ✗ No | ✗ No | ✓ Yes |
| **Indoor Farming Features** | ✓ Yes | ✓ Yes | ✗ No | ✗ No |
| **Modern Web-based UI** | ✓ Yes | ✗ No | ✓ Yes | ✗ Some |
| **Wireless IoT Network** | Yes | ✗ No | ✓ Yes | ✓ Yes |
| **Advanced Optimization Methods** | ✓ Machine Learning | ✗ No | ✗ Some | ✗ Some |
| | **Quotation (based on crop output)** | $3000 for one unit + Installation Fees | Quotation | Quotation |

# Ambient aims to provide full-coverage of the Building Technology Ecosystem




**2021**

**Software Subscription**

**2022**

**2023**

**2024**

**2025**

Software Subscription: Data Visualization

Software Subscription: AI-enabled Optimization

Software Subscription: Controls/Automation

Sensor Assembly

Sensor Integration

Sensor Manufacturing

Controls Assembly

Controls Integration

Controls Manufacturing (OEM)

**OEM Market**



**Building Technology**

